HW 4: DDL Jun. 1, 2026

Python Programming

1. Lab # 7 (4 Pts)
2. Lab # 8 (4 Pts)

Questions for Tech Report (6 Pts, $\leq$ 3 Pages)

1. **Gaussian mixture model with outliers.** Suppose that the observed data contains several outliers. The mixture model can be:

$$p(\boldsymbol{x}) = \sum_{k=1}^{K} w_k p(\boldsymbol{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) + w_{K+1} p_{noise}(\boldsymbol{x}), \quad \boldsymbol{w} = [w_1, ..., w_{K+1}] \in \Delta^K \tag{25}$$

   and w_{K+1} is probability that the sample is an outlier from Uniform$([0, c]^D)$, i.e., $p_{noise}(\boldsymbol{x}) = \frac{1}{c^D}$ if $\boldsymbol{x} \in [0, c]^D$. Modify the EM algorithm to learn this model. (3 Pts)

2. **Revisit mean-shift.** Derive the mean-shift as a maximum likelihood estimation method (refer to (24)). Derive a modified mean-shift as MAP if x owns a prior $\mathcal{N}(\mu, \sigma^2)$. What if the prior is a GMM $p_{\text{prior}}(x) = \frac{1}{K} \sum_k p(x; \mu_k, \sigma_k^2)$? (3 Pts)